

Every Delivery Room

Neonatal Resuscitation Training at National Scale and the Limits of Aggregate Evidence

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Abstract. Between 2000 and 2004 the Ministry of Health of El Salvador trained the perinatal staff of all 28 public delivery hospitals in the American Academy of Pediatrics Neonatal Resuscitation Program, and it kept training them every year until 2019. Neonatal mortality fell by more than half over the period, and a manuscript prepared with the program's leaders attributed a large share of that decline to the training on the strength of a before-and-after comparison. This paper rebuilds the analysis from the original Ministry of Health series, audits the data, and asks what a single national series can establish once secular trends, a change in the reporting system in 2009 and the region's own progress are taken into account. Three findings emerge. The before-and-after comparison is uninformative: the administrative series was already falling seven percent a year before 2000 and did not accelerate afterwards, and the break the earlier analysis dated to 2009 is a measurement event. Against a synthetic Latin American comparator built from UN IGME estimates, national neonatal mortality fell 13 to 24 percent more than expected over 2000-2019, and neonatal deaths kept pace with post-neonatal deaths when elsewhere in the region they did not, but placebo tests place gaps of that size within the range produced by untreated countries. The most program-specific evidence sits in the roll-out years: first-week mortality in the delivery hospitals fell 26 percent between 2000 and 2004 while mortality at 7 to 27 days did not, and deaths attributed to birth asphyxia fell 43 percent against 13 percent for all other causes. I use these patterns to bound the effect at roughly half a death per thousand institutional births, show that the program is cost-effective by any regional threshold even at a fraction of that bound, and set out the two datasets, a hospital-level panel and the country's vital registration by cause and age in days, that would turn a suggestive answer into a conclusive one, together with the household surveys that would show whether the decline reached the births the program could not: those outside a facility, the poorest and the most remote.

Keywords: neonatal mortality, resuscitation training, El Salvador, synthetic control, interrupted time series, program evaluation. **JEL:** I12, I18, O15, C23.

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1. Introduction

About one newborn in ten needs some help to start breathing, and a much smaller share, three to six percent of live births, needs ventilation with a bag and mask (Wall et al. 2009). When that help does not come, or comes late or wrong, the result is recorded as birth asphyxia or, in current usage, an intrapartum-related event. Such events account for roughly a quarter of the

world's neonatal deaths, close to 700,000 a year in the early 2010s, and for a comparable number of fresh stillbirths, as well as for lasting neurological injury among many survivors (Lawn et al. 2014; Liu et al. 2016). The skills required to prevent most of them are simple: drying, stimulation, clearing the airway, and positive-pressure ventilation within the first minute of life. They can be taught in two days to anyone who attends births, and they are the content of the Neonatal Resuscitation Program (NRP) of the American Academy of Pediatrics and of its low-resource descendant, Helping Babies Breathe. Trials and before-and-after studies in facilities from Zambia to Tanzania to Nepal have found that training birth attendants in these skills reduces early neonatal deaths, sometimes by nearly half (Msemo et al. 2013; KC et al. 2016), although the best-designed cluster-randomized trial found no additional mortality benefit from NRP once basic newborn care had been taught (Carlo et al. 2010). What remains genuinely open is a different question: what happens when a government trains an entire country's delivery staff, keeps retraining them for two decades, and the only record of the outcome is a national series compiled by its own ministry.

El Salvador offers an unusually clean case of that kind. Starting in 2000, its Ministry of Health, through a National Neonatal Resuscitation Committee and with the American Academy of Pediatrics, the Pan American Health Organization and USAID, ran instructor courses and then trained the perinatal staff of all 28 public delivery hospitals. Coverage of eligible providers reached three quarters by 2004, refresher courses continued every year until the pandemic interrupted them in 2020, and in total the program delivered close to 18,000 trainings. Over the same period the neonatal mortality rate reported by the Ministry fell from about 11 per 1,000 live births to about 5, and the national estimate of the UN Inter-agency Group for Child Mortality Estimation (UN IGME) fell from 15 to 6.5. A manuscript prepared with the program's leaders, on which I served as the analyst, read this record as evidence of a large and persistent effect of the training, on the basis of a comparison of the years with training against the years without (Revelo-Castro et al. 2023).

Who the program reached matters for how its results should be read. The 28 hospitals belong to the Ministry of Health, on which roughly four fifths of the population depends for care; formal-sector workers and their families are covered by the social security institute, and private hospitals serve a small minority. The households whose newborns are most likely to die of an intrapartum event, poor, rural and far from a referral hospital, are the households that give birth in Ministry facilities, and the program made no selection among them: the objective was a trained person at every delivery, whoever the mother was. That is the sense in which a cheap, universal, public-sector intervention of this kind is an instrument of equity, and it is the reason the cost-effectiveness bounds in Section 7 are worth stating with care. The same design has a boundary. In 2000, somewhere between a tenth and a third of births in El Salvador took place outside a facility, depending on whether attendance by trained midwives is counted, and those births, almost certainly the poorest and the most remote, were beyond the reach of a program that acts in delivery rooms. A hospital-based program can lower the national average and, for as long as institutional delivery is incomplete, widen the gap between births inside and outside the system. The aggregate data used here cannot see that gap, and the paper does not pretend otherwise; Section 8 names the surveys that can.

This paper takes that manuscript apart and asks what its data can actually establish. The exercise matters beyond El Salvador. Ministries and professional societies routinely evaluate national programs from the same kind of material, one aggregate series and a start date, and the temptation to read a downward trend as a program effect is strong precisely when the program is one that clinical evidence says should work. The question is not whether resuscitation training saves lives in a delivery room, which the trial literature has largely settled, but whether a national series can tell us how many, and what would have to be true of the data for it to do so.

I proceed in four steps. The first is an audit of the data as received: two Ministry spreadsheets, a study master file and a Stata driver. The audit identifies a transcription discrepancy, a set of imputed pre-program values that the earlier analysis had treated as observations, and a discontinuity in 2009 that affects every rate except the one least sensitive to what happens after a newborn leaves the hospital. The discontinuity coincides with the launch of the 2009 health reform and with the start of the Ministry's online morbidity and mortality reporting system, and I treat it as a change in measurement rather than as an outcome. The second step re-estimates the before-and-after comparison as a segmented interrupted time series with a pre-program trend. The third places El Salvador against a synthetic comparator constructed from 24 Latin American and Caribbean countries using UN IGME estimates, with permutation inference, an augmented estimator, alternative data vintages and donor pools, and an in-time placebo. The fourth exploits the structure of the intervention itself: resuscitation acts in the first minutes of life, so its effects should concentrate among first-week deaths and be absent from deaths after the first month, and they should show up in deaths attributed to asphyxia rather than in deaths from prematurity, infection or malformation. I use the age and cause composition of neonatal deaths, within El Salvador and relative to the region, as the most program-specific evidence the data contain, in the spirit of the cause-of-death designs that economists have used to isolate the contribution of medical technologies to mortality declines (Jayachandran, Lleras-Muney and Smith 2010; Alsan and Goldin 2019).

Three results follow. First, the before-and-after comparison cannot be used. The administrative neonatal mortality rate was falling at seven percent a year in 1995-1999, and the slope did not steepen after 2000; the earlier manuscript's coefficient measures the trend and the 2009 measurement break. Second, the national comparison is suggestive. Against its synthetic counterpart El Salvador's neonatal mortality fell 13 to 24 percent more than expected between 2000 and 2019, depending on the estimator and the data vintage, with a gap that widens over time; and the ratio of neonatal to post-neonatal mortality, which nets out improvements common to all infant deaths, stayed flat in El Salvador while it rose in the synthetic control. Placebo tests place these gaps at rank 8 or 9 among 25 countries, so they are consistent with a program effect and with any other improvement in newborn care in El Salvador after 2000. Third, the roll-out years look the way a resuscitation program should. In the delivery hospitals, first-week mortality fell 26 percent between 2000 and 2004 while mortality at 7 to 27 days rose 13 percent and post-neonatal mortality fell 17 percent; the first-week share of neonatal deaths dropped from 57 to 47 percent; and institutional deaths attributed to birth asphyxia fell 43 percent between 1999 and 2004 against 13 percent for all other causes, a difference of about half a death per thousand live births. These are six annual points from hospital reporting forms, and I present them as a bound rather than as an estimate.

The bound is enough to settle the policy question, if not the scientific one. At the roughly 900 trainings a year the program delivered, and at unit costs in the range reported for NRP and Helping Babies Breathe courses, the cost per disability-adjusted life year averted stays below one tenth of El Salvador's GDP per capita at the upper bound and below two thirds of it at one tenth of the upper bound. The scientific question can be settled too. The study protocols describe six-month periods per hospital before and after each hospital's training, with deaths by cause and birth weight; if that file survives, a staggered difference-in-differences across 28 hospitals would replace everything in this paper. Failing that, El Salvador's vital registration, published by the World Health Organization by cause of death and age in days, allows the cause-specific comparison to be made nationally and annually. The paper closes with that agenda.

The rest of the paper is organised as follows. Section 2 describes the program, the health system in which it operated and the manuscript this paper revisits. Section 3 reviews the evidence on resuscitation training and the methods available for evaluating national programs from aggregate data. Section 4 presents the data and the audit. Section 5 sets out the empirical strategy. Section 6 reports results, Section 7 turns them into bounds and a cost-effectiveness assessment, Section 8 describes what would settle the question, and Section 9 concludes.

2. Background

2.1 Intrapartum-related mortality and the resuscitation programs

Neonatal deaths, those in the first 28 days of life, now account for close to half of all deaths under five, and their share has risen as deaths from infection and diarrhoea after the first month have fallen faster (Lawn et al. 2014; UN IGME 2023). Within the neonatal period, three causes dominate: complications of preterm birth, intrapartum-related events, and infections. The intrapartum-related group is distinctive in two respects that matter for this paper. Its deaths are concentrated in the first hours and days of life, so they load almost entirely on the early neonatal rate (deaths at 0 to 6 days) rather than on the late neonatal rate (7 to 27 days), and they are the deaths that a person trained in resuscitation, present at the birth with a bag and mask, can prevent. A program that works should therefore change the early neonatal rate, leave the late and post-neonatal rates alone, and shift the cause composition of neonatal deaths away from asphyxia.

The NRP was created by the American Academy of Pediatrics and the American Heart Association in 1987 as a standardised curriculum with a train-the-trainer structure, and it was progressively adapted for international use after the first ILCOR consensus on newborn resuscitation in 1999 (Kattwinkel et al. 1999; Niermeyer et al. 2000). In 2010 the Academy launched Helping Babies Breathe, a one-day, simulation-based course for low-resource settings with a simplified algorithm built around the "golden minute" (Niermeyer 2015). Most of the effectiveness literature reviewed in Section 3 concerns one or the other. The Salvadoran program adopted the full NRP, in Spanish, before Helping Babies Breathe existed, and kept it throughout.

2.2 The program in El Salvador

In 1999 the Ministry of Health, as part of its commitments under the international agreements on child survival, set out a package of measures for the newborn: promotion of institutional delivery, expansion of antenatal and postpartum coverage, perinatal mortality committees in every facility, equipment for delivery and resuscitation rooms, more paediatricians and obstetricians in second- and third-level hospitals, and technical training of the staff who attend births. The resuscitation component was organised by a National Neonatal Resuscitation Committee formed by the Ministry and the Salvadoran paediatric association, with the American Academy of Pediatrics providing the curriculum and the initial faculty, the Pan American Health Organization helping to organise the instructor courses, and the USAID-funded SALSA project financing implementation (Revelo-Castro et al. 2004).

The first national instructor course took place in 2000. Instructors then ran provider courses in every one of the 28 public hospitals with a delivery room; the target population was all medical and paramedical staff involved in the direct care of the newborn: nurses, general practitioners, residents, obstetricians, paediatricians, anaesthetists and respiratory therapists. The stated objective was operational rather than educational: at least one person trained in neonatal resuscitation present at every delivery. Coverage of eligible staff rose from 5 percent in 2000 to 12 percent in 2001, 38 percent in 2002, 60 percent in 2003 and 76 percent in 2004, which, together with the cumulative counts, implies an eligible workforce of about 4,500 providers by the end of the roll-out (Revelo-Castro et al. 2005). After 2004 the program moved to a maintenance regime: new staff, residents rotating through the maternities and periodic recertification. Annual trainings never fell below 560 and averaged close to 900; by 2019 the cumulative total was 17,695, several times the size of the workforce, which is what a refresher policy in a system with staff turnover produces. The Committee recorded trainings monthly, and in compiling the study data it applied a three-month lag after each course before counting its participants as trained, on the reasoning that skills take time to be put into practice. No courses were held in 2020.

Two features of this design shape what can be learned from it. The roll-out was national and fast, so there is no untreated region of El Salvador to compare with; and it was continuous, so there is no clean “off” period either. The evaluation problem is one of a single treated unit observed over time, the setting for which interrupted time series and synthetic control methods were developed.

2.3 The health system, 1995-2020

The program operated in a health system that was changing on several fronts. Institutional delivery, already high by regional standards, rose further: World Development Indicators put births attended by skilled staff at 87 percent in 1993, 92 percent in 2003 and above 97 percent from 2009, and the Ministry’s own account describes hospital delivery care of about 95 percent by the late 2000s.¹ Total fertility fell, adolescent fertility fell from 113 to 79 births per thousand

¹ The two measures differ in definition: the WDI series counts attendance by trained personnel, including trained midwives outside facilities, while the Ministry’s figure refers to hospital deliveries. Both point to the same conclusion,

women aged 15 to 19 between 1995 and 2010, the urban share of the population rose from 52 to 65 percent, and real GDP per capita grew by about 40 percent between 2000 and 2019, slowly by the standards of the region but steadily outside the crisis years of 2009. Under-five mortality, in the UN IGME estimates, fell from 63 per thousand in 1990 to 34 in 2000 and 13 in 2019.

The most consequential change for this paper came in 2009. The government that took office in June of that year launched a health reform that eliminated the remaining user fees in Ministry facilities, created community health teams (ECOS Familiares) from 2010, expanded the hospital network and, on the information side, introduced an online morbidity and mortality reporting system (SIMMOW) whose published series begin in 2009 (MINSAL 2020). Section 4 shows that the Ministry's mortality series break sharply in that year, in a pattern that is characteristic of a change in the capture of deaths rather than of a change in the risk of dying. The reform is also a competing explanation for any improvement in newborn survival after 2009, which is one reason to weight the evidence from the roll-out years more heavily than the evidence from the later period.

2.4 The manuscript this paper revisits

The study team first reported results at the Pediatric Academic Societies meeting in 2005, when the roll-out was nearly complete: all-cause neonatal mortality in the hospital reporting forms had fallen from 12.2 to 10.2 per thousand between 1999 and 2004 and mortality attributed to birth asphyxia from 1.22 to 0.70 (Revelo-Castro et al. 2005). The abstract closed with the appropriate caveat that controlled trials were needed because of possible confounders. A manuscript prepared between 2020 and 2023 extended the series to 2020 and added three statistical procedures: supremum Wald tests for structural breaks in each mortality rate, sign tests of the direction of change, and an ordinary least squares regression of log mortality on an indicator equal to one in the training years 2000-2019 and zero in 1995-1999 and 2020. The regression coefficients, between minus 0.30 and minus 0.94 log points, were read as the proportional reductions in mortality associated with the training.

That reading does not survive scrutiny, for reasons that Section 5 sets out formally. The indicator compares five early years with twenty later ones in a series that was falling throughout, so the coefficient is a trend, not a treatment effect. Two of the four outcomes had no pre-2000 observations at all; the driver file imputed them by regression on the third and used the imputations as the comparison period. The year 2020 was coded as untreated although the trained workforce was still in place. And the structural breaks that the manuscript attributed to the decline in annual trainings after 2009 coincide with the change in the reporting system. None of this reflects on the program, and the study team's substantive hypothesis may well be right. It reflects the limits of the design, which this paper tries to make explicit and, where possible, to overcome.

that by the 2000s the large majority of births in El Salvador took place where the program could reach them, and that the share kept rising.

3. What is known, and how it is known

Two bodies of work bear on this paper. The first asks whether training birth attendants in resuscitation saves newborn lives, and if so how many; it is clinical and epidemiological, and its designs range from cluster-randomized trials to single-hospital before-and-after comparisons. The second asks how to learn about the effect of a policy that hits one unit at one moment, with no experimental control; it is econometric, and it has developed rapidly over the past fifteen years. The two literatures rarely meet. National resuscitation programs have been evaluated almost exclusively with the tools of the first, and the tools of the second have rarely been applied to a health intervention whose clinical mechanism is as well understood as this one. This section reviews both, and draws from the intersection the design used in the rest of the paper.

3.1 Does resuscitation training reduce neonatal mortality?

The clinical logic is not in doubt. The techniques taught in NRP restore breathing in the overwhelming majority of newborns who need help, and observational work in delivery rooms has shown that the timing of the first ventilation, within the first minute, is what separates survival from death among non-breathing infants (Ersdal et al. 2012). The empirical question has always been whether training changes what happens at births, and whether that change is large enough to move mortality rates measured at the level of a hospital, a district or a country.

The early evidence came from before-and-after comparisons in Asia. A single hospital in Zhuhai reported that first-week mortality fell from 9.9 to 3.4 per thousand after NRP was introduced (Zhu et al. 1997); a network of Indian teaching hospitals reported a fall in asphyxia-related deaths and in hypoxic-ischaemic encephalopathy after training (Deorari et al. 2001); and a rural Indian program reported that asphyxia mortality fell by more than half within two years (Tholpadi et al. 2002). The authors of these studies were candid about historical controls and inconsistent definitions, and the systematic reviews that later pooled them rated the evidence as low quality while finding it consistent in direction: training was associated with reductions in intrapartum-related and early neonatal deaths in facilities of the order of 30 percent, the figure adopted by the Lives Saved Tool after a meta-analysis and a Delphi exercise (Lee et al. 2011; Pammi et al. 2016).

The strongest design in the literature gave a more sobering answer. The First Breath trial trained birth attendants in 96 rural communities of six countries, first in the World Health Organization's Essential Newborn Care course and then, in a cluster-randomized second phase, in NRP. Essential newborn care training was followed by a fall in stillbirths (a relative risk of 0.69) and no change in seven-day mortality (a relative risk of 0.99); the subsequent NRP training produced no further reduction in seven-day mortality, stillbirths or perinatal mortality relative to control clusters (Carlo et al. 2010). The result is usually read in two ways that are both relevant here. Births in the trial were attended mostly by traditional birth attendants at home, so the incremental content of NRP over basic newborn care, essentially bag-and-mask ventilation, may have been rarely used; and the trial measured mortality within seven days in communities where many deaths occur for reasons that resuscitation cannot touch. The result is a reminder that the effect of training depends on who is trained, on where births happen and on what the trainees could already do.

Helping Babies Breathe, launched in 2010 for exactly the settings where First Breath found little effect, generated a second wave of evidence, again mostly before-and-after. The Tanzanian implementation study in eight hospitals reported a 47 percent reduction in early neonatal mortality and a 24 percent reduction in fresh stillbirths after training, together with more stimulation and less suctioning and ventilation at birth (Msemo et al. 2013). In Belgaum, India, training reduced fresh stillbirths but not seven-day mortality (Goudar et al. 2013). In a tertiary hospital in Nepal, training combined with a quality-improvement cycle roughly halved intrapartum stillbirths and first-day deaths (KC et al. 2016), while a subsequent stepped-wedge trial that scaled a similar package to twelve public hospitals found weaker effects, a reminder that implementation is part of the treatment (KC et al. 2019). A two-country pre-post study of facility birth attendants found no measurable change in perinatal mortality (Bellad et al. 2016). Pooled across studies, the reviews find reductions in early neonatal mortality and intrapartum-related stillbirth that are meaningful in size and fragile in design (Dol et al. 2018; Versantvoort et al. 2020; Patel et al. 2017).

A separate strand concerns the durability of the skills. Simulated performance after a one-day course improves and then decays within months, and it does not translate automatically into clinical management (Ersdal et al. 2013; Bang et al. 2016). Frequent, brief, on-site simulation practice appears to be what sustains both the skills and the mortality gains (Mduma et al. 2015), and the quality-improvement components of the Nepal work point in the same direction. This literature has a direct implication for El Salvador. A program that trained three quarters of the workforce in four years and then stopped would be expected to show an effect that fades; a program that kept delivering close to 900 trainings a year for another fifteen years, in a workforce of about 4,500, is closer to the refresher regime that the skill-retention studies recommend. The annual flow of trainings after 2004 is not a dose in the usual sense. It is maintenance.

Two gaps in this literature frame the contribution of the present paper. First, almost all of it concerns facilities or districts observed for one to three years around a training event, and almost none of it concerns a country observed for twenty. The exceptions are implementation and cost studies of the national Helping Babies Breathe roll-out in Tanzania, which document what scale-up requires without attempting a national mortality estimate (Arlington et al. 2017; Chaudhury et al. 2016). Second, the Spanish-language NRP has been taught across Latin America since the 1990s, and the region's neonatal mortality has fallen by more than half since 2000, yet published effectiveness evidence from Latin America is close to nonexistent. El Salvador's program is one of the few with a documented national roll-out, a clean start date and two decades of continuous training records.

3.2 Learning from one treated unit

The evaluation problem in this paper is the one that arises whenever a policy is adopted by a whole country at once. There is one treated unit, a long pre-period, and a post-period during which everything else also changed. Three families of methods address it, and the paper uses all three.

Interrupted time series regression is the workhorse of health-policy evaluation with aggregate data. A segmented model with a level and a slope change at the intervention date, autocorrelation-robust standard errors and placebo interruption dates gives a transparent test of whether the series departed from its pre-intervention trajectory (Bernal, Cummins and Gasparrini 2017; Kontopantelis et al. 2015). Its credibility rests on the pre-period being long enough to estimate the counterfactual trend and on nothing else changing at the same date; with five pre-period observations, as here, the first condition is barely met, and the method serves better as a description of the series than as an estimator.

The synthetic control method replaces the extrapolated trend with a data-driven comparison: a weighted average of untreated units chosen so that it reproduces the treated unit's pre-intervention path, with weights constrained to be non-negative and to sum to one (Abadie and Gardeazabal 2003; Abadie, Diamond and Hainmueller 2010). Its attractions are that the counterfactual is explicit, that the weights can be inspected, and that inference can proceed by permutation, re-estimating the model with each control unit in the role of the treated one and asking where the actual estimate falls in the resulting distribution (Abadie, Diamond and Hainmueller 2015). Its limits are equally well understood. When the pre-period fit is imperfect the estimate is biased (Ferman and Pinto 2021); when the treated unit lies outside the convex hull of the donors no set of weights will fit it; and when all pre-period outcomes are used as predictors the covariates play no role in the weights (Kaul et al. 2022). Recent work relaxes the original constraints in different directions. The augmented synthetic control adds an outcome model that corrects the remaining pre-period imbalance (Ben-Michael, Feller and Rothstein 2021), synthetic difference-in-differences combines unit and time weights (Arkhangelsky et al. 2021), and conformal methods provide inference that does not rely on the permutation being exchangeable (Chernozhukov, Wüthrich and Zhu 2021). Applications to population health are now numerous enough to have been reviewed, with the recurring caution that outcome series which are themselves modelled, as national mortality estimates are, make pre-period fit easy to achieve and permutation ranks the only honest summary of uncertainty (Bouttell et al. 2018; Kreif et al. 2016). Abadie's (2021) survey of feasibility and data requirements is the reference point for the choices made in Section 5.

The third family is older than either and comes from the economics of mortality decline. When a medical technology acts on some causes of death and not on others, the untreated causes provide a within-unit control for everything that moves all mortality together. Jayachandran, Lleras-Muney and Smith (2010) used the arrival of sulfa drugs in 1937 in this way, comparing the causes they treated (pneumonia, scarlet fever, maternal sepsis) with the causes they did not, and Alsan and Goldin (2019) used the same logic to isolate the effect of water and sewerage infrastructure on child deaths from waterborne disease; Cutler and Miller (2005) and Anderson, Charles and Rees (2022) apply variants of it to the public-health interventions of the early twentieth century. The design has an obvious counterpart here. Resuscitation acts on intrapartum-related deaths, which cluster in the first week, and not on deaths after the first month; and it acts on asphyxia, not on prematurity, infection or malformation. The age and cause composition of neonatal deaths therefore carries information about the program that the all-cause rate does not, and it can be combined with the cross-country comparison in a triple difference: El Salvador against the region, treated ages or causes against untreated ones, after

against before. The same logic underlies the economics literature on the returns to intensive newborn care, which identifies effects from thresholds and timing rather than from aggregate trends (Almond et al. 2010; Bharadwaj, Løken and Neilson 2013), and it is the part of the design in this paper that is most specific to the intervention.

3.3 What the region's own decline implies

Any comparison across Latin America has to reckon with the fact that neonatal mortality fell almost everywhere in the region after 1990, for reasons that have little to do with resuscitation. Fertility declined, incomes and women's schooling rose, delivery moved into facilities, and the share of infant deaths occurring in the neonatal period rose as post-neonatal deaths from infection collapsed (Cutler, Deaton and Lleras-Muney 2006; Lawn et al. 2014). In the UN IGME estimates used below, the average neonatal mortality rate of the 24 comparison countries fell from 21 per thousand in 1990 to 16 in 2000 and 10 in 2019. El Salvador's rate fell faster than the regional average in every decade, including the 1990s, before the program existed. A comparison that does not account for that is a comparison of El Salvador's secular trajectory with everyone else's, and it is the reason the paper builds a synthetic comparator from countries that matched El Salvador's 1990s path rather than from the regional mean. It is also the reason the composition evidence is given as much weight as the cross-country evidence: a country that is doing many things right will show a faster decline in all-cause mortality, but only a resuscitation program should change the age and cause profile of the deaths that remain.

4. Data

The analysis combines two kinds of data that measure different populations and must not be mixed: the administrative series compiled by the Ministry of Health of El Salvador for the study, and the cross-country estimates produced by UN IGME and the World Health Organization. Table 1 lists them. Every transformation is documented in a provenance file that accompanies the code, and the audit in Section 4.2 lists the decisions that differ from the 2023 manuscript.

Table 1. Data sources

Series	Source and vintage	Years	Population	Use
Neonatal mortality (0-27 days)	MINSAL, transmitted to the study team	1995-2020	Institutional (facility births of 500 g or more, excluding lethal malformations)	Interrupted time series; composition
Infant and early neonatal mortality (0-364 and 0-6 days)	MINSAL	2000-2020	Institutional	Composition; derived late (7-27 d) and post-neonatal (28-364 d) rates
Providers trained in NRP	Comité Nacional de Reanimación Neonatal	2000-2019	All 28 public delivery hospitals	Roll-out timing; training stock
Institutional neonatal and asphyxia mortality	Revelo-Castro et al. (2005), hospital reporting forms	1999-2004	Institutional	Cause composition during the roll-out
Neonatal, infant and under-five mortality, all countries	UN IGME, 2021 vintage (Our World in Data distribution)	1990-2020	National	Synthetic control; control outcome

Series	Source and vintage	Years	Population	Use
Neonatal mortality, all countries	UN IGME, 2024 vintage (Gapminder distribution of childmortality.org)	1990-2024	National	Robustness
Neonatal deaths by cause	WHO Global Health Observatory, MCEE estimates	2000-2017	National	Cause composition (found uninformative)
GDP per capita, urban share, women's schooling, skilled attendance, adolescent fertility, health spending	World Bank WDI, IHME, WHO GHED (Gapminder distribution)	1990-1999 means	National	Synthetic control predictors

4.1 The administrative series

The Ministry transmitted two spreadsheets. The first, prepared for the 2020 draft, contains infant, neonatal and early neonatal mortality per thousand live births for 2000-2018 and the number of providers trained in each year; the second updates the mortality rates through 2020. Where the two overlap they agree exactly. Neonatal mortality for 1995-1999 was transmitted separately and is recorded in the study's master file and in the 2020 draft as the Ministry's national figures for those years; its 1999 value (12.63) is close to both the 2000 value of the later series (11.38) and the 1999 value in the hospital reporting forms summarised for PAS 2005 (12.2), so the two segments are treated as one series, with the caveat that their compilation may have differed. The rates refer to institutional births: the protocols define the population as live births of 500 grams or more in the 28 public hospitals with delivery rooms, excluding infants with known lethal malformations and infants brought to a hospital after delivery elsewhere. Deaths are classified by age at death into early neonatal (0-6 days), late neonatal (7-27 days), neonatal (0-27 days) and post-neonatal (28-364 days); the late and post-neonatal rates are obtained by subtraction.

Table 2 reports the series at the years that organise the analysis. Two features stand out before any modelling. Neonatal mortality fell by a third between 1995 and 1999, before the program, and by another third between 2000 and 2008, at a somewhat slower annual pace. And between 2008 and 2009 every rate except the early neonatal rate dropped by an amount that has no counterpart anywhere else in the series: neonatal mortality fell 39 percent, late neonatal mortality 76 percent, post-neonatal mortality 34 percent and infant mortality 37 percent in a single year, while early neonatal mortality fell 12 percent. After 2009 the late and post-neonatal rates drift back up, by 69 and 36 percent respectively over the following decade, while the early neonatal rate is flat. Section 4.5 returns to this pattern.

Table 2. Ministry of Health series at selected years (deaths per 1,000 live births) and providers trained

Year	Neonatal (0-27 d)	Early neonatal (0-6 d)	Late neonatal (7-27 d)	Post-neonatal (28-364 d)	Infant	Trained in year	UN IGME neonatal (national)
1995	17.00	.	.	.	.	0	18.9
1999	12.63	.	.	.	.	0	15.9
2000	11.38	6.5	4.88	4.72	16.10	239	15.0
2004	10.30	4.8	5.50	3.90	14.20	1,349	12.1
2008	7.36	4.2	3.16	4.00	11.36	1,342	10.1
2009	4.47	3.7	0.77	2.65	7.12	1,230	9.7

Year	Neonatal (0-27 d)	Early neonatal (0-6 d)	Late neonatal (7-27 d)	Post-neonatal (28-364 d)	Infant	Trained in year	UN IGME neonatal (national)
2014	6.40	4.9	1.50	3.30	9.70	760	7.8
2019	5.10	3.8	1.30	3.60	8.70	883	6.5
2020	5.00	3.8	1.20	2.90	7.90	0	6.2

Note: Institutional rates from MINSAL; late and post-neonatal rates by subtraction; the last column is the UN IGME 2021 national estimate for comparison. Infant and early neonatal rates are not observed before 2000.

4.2 Audit of the study files

Rebuilding the data from the spreadsheets rather than from the master file produced five corrections to the 2023 manuscript's inputs.

1. *A transcription error.* The number of providers trained in 2011 is 684 in the Ministry spreadsheet and 648 in the master file; the five-year block total printed in the spreadsheet (3,599 for 2010-2014) is only consistent with 684, which is used here.
2. *Imputed pre-program values.* The Stata driver regressed infant and early neonatal mortality on neonatal mortality over 2000-2020, predicted the two series for 1995-1999, replaced the missing cells with the predictions and then estimated treatment effects on the completed series. The imputed cells are dropped. Pre-program comparisons are possible only for the 28-day rate.
3. *The year 2020.* The manuscript coded 2020 as untreated because no courses were held. The trained workforce was still in place, and 2020 is also the first pandemic year; it is treated as a post-program year affected by COVID-19 and excluded from the main estimation windows.
4. *Two versions of 1999-2004.* The PAS 2005 abstract reports institutional neonatal mortality from the hospital reporting forms of 12.2, 11.1, 11.9, 11.9, 10.5 and 10.2 for 1999-2004; the spreadsheet reports 12.63, 11.38, 11.20, 10.90, 10.40 and 10.30. The compilations differ in level by a few tenths and in year-to-year movement. The spreadsheet series is used as the main series; the abstract series is the only source of cause-specific rates and is used for that purpose.
5. *Roll-out coverage.* The cumulative counts of trainees for 2000-2004 (239, 538, 1,699, 2,721 and 3,415) correspond to 5, 12, 38, 60 and 76 percent of eligible staff in the 2005 presentation, which implies an eligible workforce of about 4,500. The 2004 flow in the 2005 table (694) is a partial-year count; the spreadsheet's full-year figure (1,349) is used.

4.3 Training flows, coverage and stock

The annual flow of trainings is the only measure of program intensity available. Three transformations are used descriptively: the cumulative count, a coverage share for the roll-out years, and a depreciated stock that discounts past trainings at 20 or 33 percent a year to reflect turnover and skill decay. None of them enters a causal estimate. With twenty annual observations and a stock that saturates by 2004, the flow after the roll-out is best understood as the maintenance regime that the skill-retention literature recommends, and the paper dates the treatment by the roll-out rather than by the flow.

4.4 The cross-country panel

The UN IGME estimates are the standard for international comparison of child mortality, and they are the only source that covers every country in the region annually since 1990. They are not observations. For a country like El Salvador they are the output of a Bayesian B-spline bias-reduction model fitted to the household surveys (FESAL 1998, 2002-03 and 2008; MICS 2014) and vital registration, and they are smooth by construction (Alkema and New 2014). The 2021

vintage is used as the main panel because it carries neonatal and infant mortality from the same model run, which the control-outcome design in Section 5.4 requires; the 2024 vintage, available for neonatal mortality only, extends the series to 2024 and is used as a robustness check. The two vintages differ for El Salvador in ways that matter for the size of the estimates: the 2024 revision places the country's recent neonatal mortality lower (4.4 per thousand in 2024 against 6.2 in 2020 in the 2021 vintage), and Section 6 reports both.

The donor pool consists of the 24 other sovereign countries of Latin America and the Caribbean with UN IGME series: Argentina, Belize, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, the Dominican Republic, Ecuador, Guatemala, Guyana, Haiti, Honduras, Jamaica, Mexico, Nicaragua, Panama, Paraguay, Peru, Suriname, Trinidad and Tobago, Uruguay and Venezuela. In several of these countries, individual hospitals adopted NRP or Helping Babies Breathe during the period, without a documented national program. That contaminates the pool in a known direction: the synthetic control estimates the effect of systematic national implementation relative to the region's diffuse adoption, and it is biased toward zero relative to the effect of training versus no training. Two of the donors experienced large shocks in the post-period, Haiti's 2010 earthquake and Venezuela's collapse after 2014, and a restricted pool without them is reported. Pre-treatment covariates for the covariate-based specification are the 1990-1999 means of log GDP per capita, the urban share, mean years of schooling of women aged 15 to 44, the share of births attended by skilled staff, the adolescent fertility rate and the under-five mortality rate.

Cause-specific neonatal mortality comes from the WHO Global Health Observatory's child cause-of-death estimates for 2000-2017. For countries without complete vital registration these are modelled fractions applied to the UN IGME envelope, and Section 6.5 shows that for El Salvador and several donors the fractions are essentially constant over time, which makes them uninformative about the mechanism studied here.

4.5 Two populations and one discontinuity

The Ministry series and the UN IGME series measure different things, and the gap between them is informative. In 2000 the institutional neonatal mortality rate was 11.4 per thousand and the national estimate 15.0; in 2019 the figures were 5.1 and 6.5. The institutional rate is lower for three reasons that all pull in the same direction: home births, which carry higher risk and were still a fifth or more of the total in the early 2000s, are excluded; infants under 500 grams and with lethal malformations are excluded; and deaths that occur after discharge are captured only if the hospital learns of them. The last point is the key to the 2009 discontinuity. A death in the first days of life almost always occurs in the hospital where the birth took place, so the early neonatal rate is robust to how deaths after discharge are recorded. Deaths at 7 to 27 days and at 28 to 364 days increasingly occur at home or in other facilities, and their capture depends on the reporting system. A change in that system would produce exactly the pattern in Table 2: a collapse in the late and post-neonatal rates, a small movement in the early neonatal rate, and a gradual recovery as the new system matures. The Ministry's published trend series from the online morbidity and mortality system begin in 2009, the year the reform started, and I treat

2009 as a change of measurement regime rather than as an outcome.² The pre-program comparisons use 1995-2008; the later period is analysed with a level shift and is given less weight.

5. Empirical strategy

5.1 The estimand and why the before-and-after comparison fails

Let y_t denote the log of a mortality rate in year t and let the program start in $t_0 = 2000$. The object of interest is the difference between the observed series and the series that would have obtained without the program, $\tau_t = y_t - y_t^N$, for $t \geq t_0$. The 2023 manuscript estimated

$$y_t = \alpha + \beta D_t + \varepsilon_t, \quad D_t = \mathbf{1}\{2000 \leq t \leq 2019\},$$

on 1995-2020 and read β as the proportional effect of training. Under this specification y_t^N is assumed constant at its 1995-1999 mean, and β is the difference between two period means of a series that was falling throughout. If mortality declines at rate g for reasons unrelated to the program, the estimate converges to g times the difference in mean date between the two periods, roughly twelve years here, whatever the program did. The Durbin-Watson statistic of the manuscript's regression (0.34) is the residual trend announcing itself. The problem is not the sample size; it is the counterfactual.

5.2 Interrupted time series on the administrative data

The natural repair, when the only data are one series, is to estimate the counterfactual trend from the pre-period. I estimate the segmented model

$$y_t = \beta_0 + \beta_1(t - t_0) + \beta_2 P_t + \beta_3 P_t(t - t_0) + \varepsilon_t, \quad P_t = \mathbf{1}\{t \geq t_0\},$$

where β_1 is the pre-program annual trend in log points, β_2 the immediate level change in 2000 and β_3 the change in the annual slope; the implied effect after k years is $\beta_2 + \beta_3 k$. The main window is 1995-2008, which ends before the reporting regime changes. A second window, 1995-2019, adds a level shift in 2009. Standard errors are Newey-West with two lags and, as a check, from an iterated Prais-Winsten AR(1) transformation; placebo interruptions in 1997 and 1998 test whether the procedure finds “effects” where there were none. With five pre-period observations the test has little power and the trend estimate is fragile, and the specification is reported for what it is: a description of whether the series bent at 2000, with the direction of the answer more informative than its p-values. The manuscript's specification is also re-estimated for reference.

² The interpretation is a hypothesis about the data system that the Ministry can confirm or reject, and the paper is written so that either answer leaves its conclusions intact. If 2009 were a genuine mortality event, it would be one that fell on deaths after the first week, which is where a resuscitation program has the least to do, and it would coincide with the elimination of user fees and the reform rather than with any change in training.

5.3 Synthetic control on national estimates

The second strategy replaces the extrapolated trend with the experience of comparable countries. Let Y_{jt} be log neonatal mortality for country j , with $j = 1$ El Salvador and $j = 2, \dots, J + 1$ the donors. The synthetic control is $\hat{Y}_{1t}^N = \sum_{j \geq 2} w_j Y_{jt}$, with weights $w_j \geq 0$, $\sum_j w_j = 1$ chosen to minimise

$$\|X_1 - X_0 w\|_V^2 = (X_1 - X_0 w)' V (X_1 - X_0 w),$$

where X_1 and X_0 collect pre-treatment predictors for El Salvador and the donors and V is a diagonal weighting matrix. Three predictor sets are used. Specification A uses every pre-period value of the outcome, 1990-1999, with uniform V ; the fit is then close to exact and covariates are irrelevant by construction (Kaul et al. 2022), which makes A the transparent benchmark. Specification B uses the outcome in 1990, 1995 and 1999 together with the six pre-period covariates, with V chosen by nested optimisation to minimise the pre-period mean squared prediction error. Specification C is the ridge-augmented estimator of Ben-Michael, Feller and Rothstein (2021), which adds to the specification-A path an outcome-model correction for whatever pre-period imbalance the weights leave, with the ridge penalty chosen by leave-one-out cross-validation among donors. The estimated effect is the post-period gap $\hat{\tau}_t = Y_{1t} - \hat{Y}_{1t}^N$, summarised as the mean over 2000-2004 (the roll-out), 2005-2009, 2010-2019 and 2000-2019, and converted from log points to percentages.

Inference is by permutation. Each donor is treated in turn as if it were the treated unit, with the remaining 23 countries as its donors, and the distribution of the resulting placebo gaps is the reference against which El Salvador's gap is judged (Abadie, Diamond and Hainmueller 2010). Two summaries are reported: the rank of El Salvador's ratio of post-period to pre-period root mean squared prediction error, which rewards units whose post-period departure is large relative to how well they were fitted, and the rank of the post-period mean gap itself. Because the UN IGME series are smooth, pre-period fit is nearly exact for many units and the RMSPE ratio becomes sensitive to tiny differences in denominators; the mean-gap rank and a version of the ratio test restricted to placebos whose pre-period fit is within five times El Salvador's guard against that. Leave-one-out estimates drop each donor with positive weight in turn. An in-time placebo assigns a fictitious treatment in 1995, fitted on 1990-1994, and asks whether the method manufactures an effect in the second half of the pre-period. Estimates in levels rather than logs, on the restricted donor pool and on the 2024 vintage complete the set.

Two remarks on interpretation. First, donor contamination biases the estimate toward zero, as noted in Section 4.4, so the synthetic control gap is a lower bound on the effect of training versus no training and an estimate of the effect of systematic national implementation versus the regional norm. Second, the gap is an estimate of every change in El Salvador after 2000 that affected newborn survival, of which the program is one component. The composition designs are what separate the program from the rest.

5.4 Age composition and a control outcome

Resuscitation acts at birth. Its effect on mortality should be concentrated among deaths in the first days of life, it should be small at 7 to 27 days, and it should be absent after the first month, when infants die of infection, malformation and the sequelae of prematurity. Post-neonatal mortality is therefore a control outcome: it responds to income, sanitation, vaccination, fertility and facility delivery, to the health reform and to the reporting system, and it does not respond to a bag and mask. Two uses follow.

Within El Salvador, the Ministry series allow the early neonatal, late neonatal and post-neonatal rates to be followed through the roll-out. A program effect should appear as a fall in the early neonatal rate during 2000-2004 that is not matched by the late neonatal rate, and as a fall in the early neonatal share of neonatal deaths. This is a comparison of six annual points with no external control, and it is reported as such.

Across countries, the ratio of neonatal to post-neonatal mortality, $r_{jt} = \ln(\text{NMR}_{jt}/\text{PNMR}_{jt})$, differences out whatever moves all infant deaths in country j in year t . Applying the synthetic control to r_{jt} yields a triple-difference estimate,

$$\hat{\delta}_t = r_{1t} - \sum_{j \geq 2} w_j r_{jt},$$

in which the weights reproduce El Salvador's pre-period ratio path. The identifying assumption is that, absent the program, El Salvador's neonatal deaths would have fallen relative to its post-neonatal deaths at the same pace as in the synthetic control, which is weaker than the assumption behind the all-cause comparison because country-specific improvements that affect all infants cancel. The raw triple difference, without weights, is reported alongside. Both are estimated on the 2021 vintage, the only one in which neonatal and infant mortality come from the same model run.

5.5 Cause of death

The most direct test is whether deaths attributed to asphyxia fell faster than deaths from other causes. The hospital reporting forms summarised for PAS 2005 give institutional neonatal mortality and asphyxia-specific mortality for 1999-2004, and I compare the change in the asphyxia rate with the change in the residual (all other causes) over the roll-out. This is the cause-specific analogue of the age-composition test and, like it, it has no external control. The WHO cause-of-death estimates would provide the cross-country counterpart if they carried independent information about El Salvador's cause fractions; Section 6.5 shows they do not.

5.6 What is not estimated

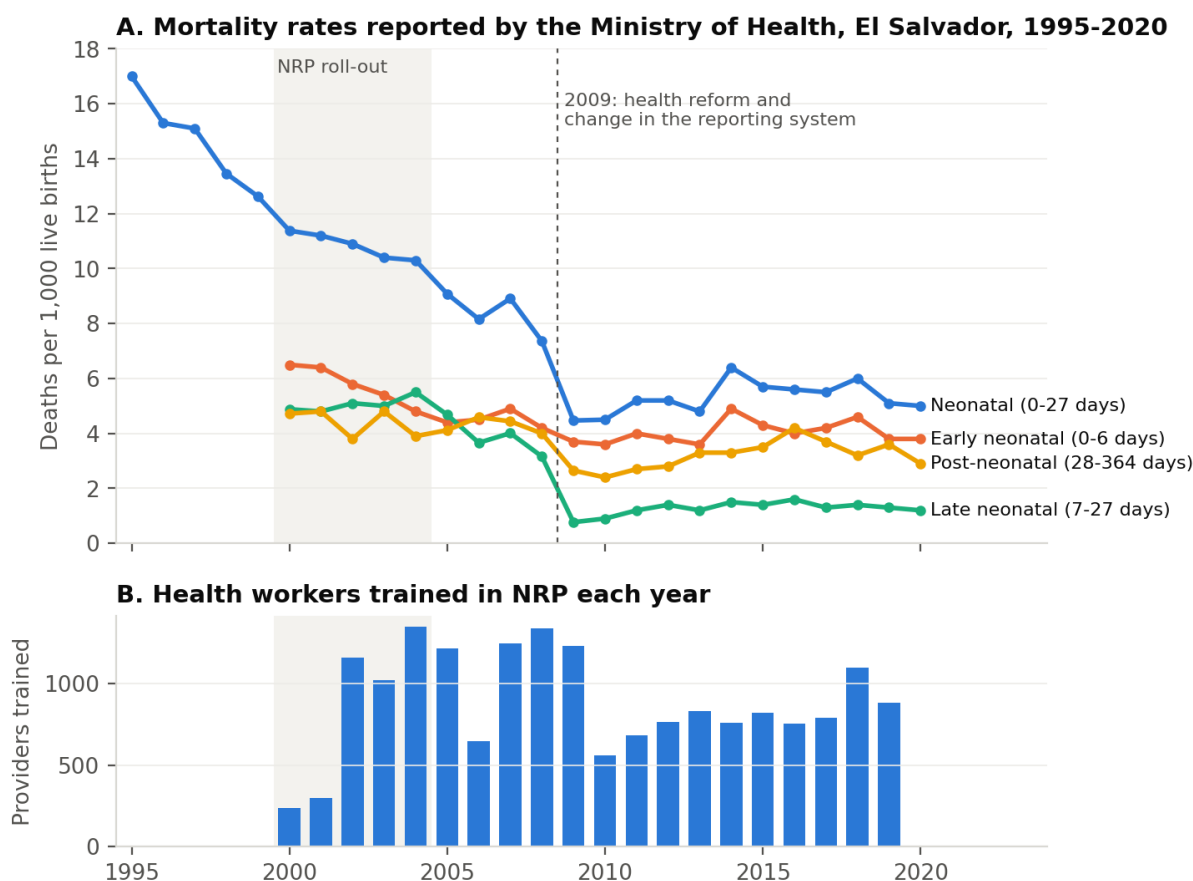
Two elements of the 2023 manuscript are dropped from the inferential apparatus. Regressions of mortality on the number of providers trained, or on a depreciated stock, are kept only as descriptive scatter plots: the flow saturates in 2004, twenty annual observations cannot identify a dose-response relationship, and heteroskedasticity-robust standard errors on series that short are not meaningful. The supremum Wald tests for structural breaks are re-run and reported in

the appendix, where they locate the 2009 break in every rate except the early neonatal one; they describe the reporting change and say nothing about the program.

6. Results

6.1 The administrative series and the interrupted time series

Figure 1 plots the Ministry series and the annual trainings. Three things are visible without any estimation: the neonatal rate was already falling steeply before 2000; the four rates move together until 2008; and in 2009 everything except the early neonatal rate drops and then drifts back.



Sources: MINSAL El Salvador; Comité Nacional de Reanimación Neonatal. Infant and early-neonatal rates are only available from 2000.

Figure 1. Mortality rates reported by the Ministry of Health, El Salvador, 1995-2020, and providers trained in NRP each year. Sources: MINSAL; Comité Nacional de Reanimación Neonatal. Infant and early neonatal rates are only available from 2000.

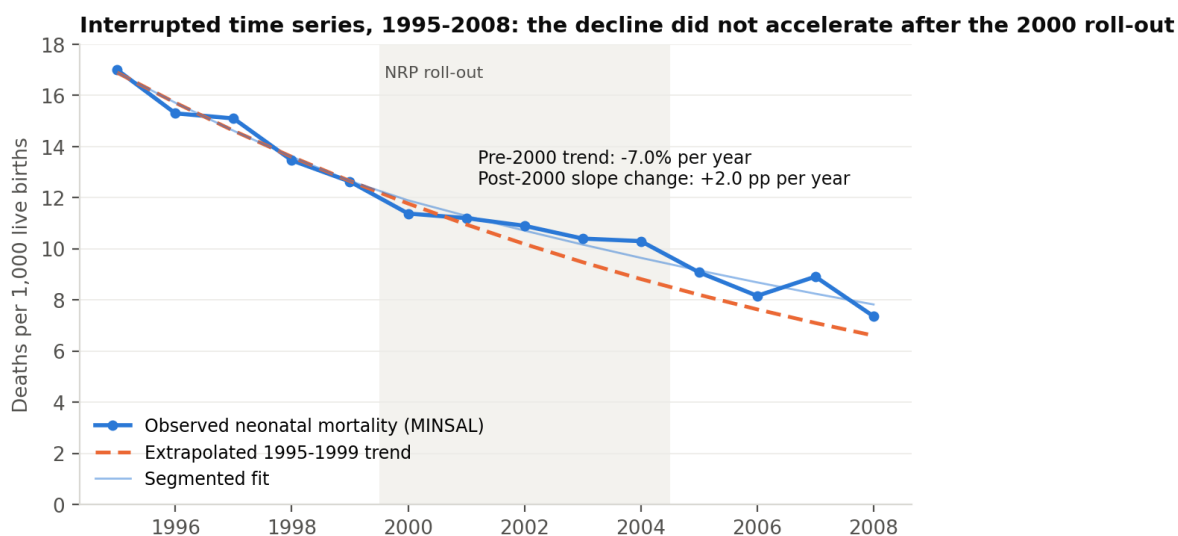
Table 3 reports the segmented regressions. In the main window, 1995-2008, the pre-program trend is minus 0.072 log points a year, a decline of 7.0 percent annually. The level change in 2000 is essentially zero (+1.1 percent) and the slope change is positive: after 2000 the rate fell 2.0 percentage points a year more slowly than before. The implied cumulative “effect” after four

years is +9 percent, that is, the observed rate in 2004 sits above the extrapolated pre-program trend, as Figure 2 shows. Extending the window to 2019 with a 2009 level shift does not change the picture for the roll-out years; the 2009 shift itself is minus 0.55 log points, a 42 percent drop, and the post-2000 slope becomes even less negative because the post-2009 series rises. Without the 2009 dummy the discontinuity loads on the slope terms and the fit deteriorates (Durbin-Watson 0.66). The manuscript's own specification, re-estimated on the corrected data, reproduces its coefficient of minus 0.57 with a Durbin-Watson statistic of 0.34.

Table 3. Interrupted time series estimates, log neonatal mortality (MINSAL)

	(1) 1995-2008	(2) 1995-2019, 2009 shift	(3) 1995-2019, no shift	(4) Manuscript specification, 1995-2020
Pre-program trend, per year	-0.072 (0.002) [0.011]	-0.072 (0.002) [0.040]	-0.072 (0.002) [0.056]	.
Level change, 2000	0.011 (0.021) [0.042]	-0.179 (0.082) [0.125]	-0.092 (0.080) [0.169]	.
Slope change after 2000, per year	0.020 (0.005) [0.011]	0.067 (0.012) [0.044]	0.026 (0.008) [0.060]	.
Level shift, 2009	.	-0.552 (0.140) [0.111]	.	.
Training-period indicator	.	.	.	-0.568 (0.168) {0.210}
Implied change after four years	+0.090 (+9.4%)	+0.090 (+9.4%)	+0.012 (+1.2%)	.
Observations	14	25	25	26
R-squared	0.97	0.93	0.84	0.32
Durbin-Watson	2.61	0.84	0.66	0.34

Note: Newey-West standard errors with two lags in parentheses; iterated Prais-Winsten AR(1) standard errors in brackets; conventional OLS standard error in braces for column (4). The Newey-West errors in column (1) are implausibly small because five pre-period points lie almost on a line; the Prais-Winsten errors are the more honest guide, and neither should be read as a test with real power.



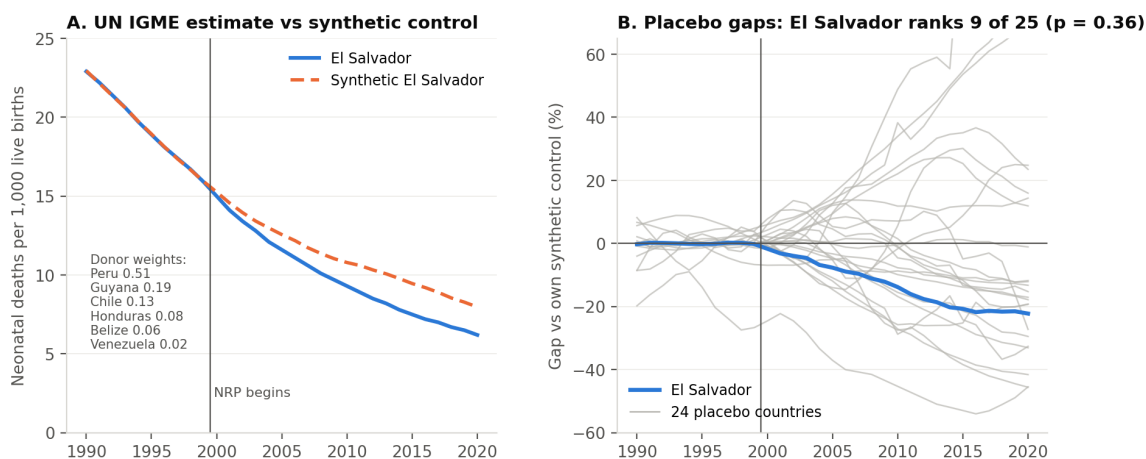
Segmented regression of log neonatal mortality with a level and slope change in 2000. Window ends in 2008, before the reporting change.

Figure 2. Interrupted time series, 1995-2008. The 1995-1999 trend extrapolated forward lies below the observed series: the decline did not accelerate after the 2000 roll-out.

The placebo interruptions make the same point from the other side. A fictitious start in 1997 yields a level change of +4 percent and a slope change of +5 points a year; a start in 1998 yields minus 5 percent and +0.6 points. A procedure that finds “effects” of this size at dates when nothing happened cannot be trusted to find a real one of similar size at 2000. The only robust conclusion from the administrative series is negative: whatever the program did to the 28-day rate, it did not bend a trend that was already steep.

6.2 Synthetic control

Figure 3 shows the main synthetic control (specification A) and its placebo distribution. Synthetic El Salvador is a combination of Peru (weight 0.51), Guyana (0.19), Chile (0.13), Honduras (0.08), Belize (0.06) and Venezuela (0.02); the pre-period fit is exact to three decimals in logs. After 2000 the two series separate gradually. The gap averages minus 4 percent during the roll-out, minus 10 percent in 2005-2009 and minus 19 percent in 2010-2019, for a mean of minus 13 percent over 2000-2019; in levels, synthetic El Salvador stands at 8.3 per thousand in 2019 against an actual 6.5, a difference of 1.8 deaths per thousand live births.



Log neonatal mortality, Latin American donor pool (24 countries), predictors = 1990-1999 outcomes. UN IGME 2021 vintage.

Figure 3. Synthetic control on UN IGME estimates (2021 vintage). Panel A: El Salvador and its synthetic control, donor weights shown. Panel B: gaps between each country and its own synthetic control; El Salvador ranks 9 of 25 on the RMSPE ratio ($p = 0.36$).

The other specifications tell the same story with different numbers (Table 4). The covariate-based specification B gives minus 14 percent over 2000-2019, with the nested optimisation placing all the weight on the 1999 outcome, as is common when a recent lag is available; the covariate balance it achieves is nonetheless informative: synthetic El Salvador matches the actual country on urbanisation and under-five mortality, is somewhat richer, has more schooled women, lower skilled attendance and lower adolescent fertility. The augmented estimator gives minus 13 percent with a ridge penalty close to zero, since there is almost no imbalance to correct. In levels the gap is 0.8 deaths per thousand on average over 2000-2019, or minus 8 percent of the synthetic level. The restricted pool without Haiti and Venezuela gives minus 14 percent. Leave-one-out estimates range from minus 7 percent (dropping Guyana) to minus 16 percent (dropping Peru). The 2024 vintage gives larger gaps, minus 24 percent over 2000-2019

and minus 47 percent over 2020-2024, because the revised series places El Salvador's recent mortality lower; Figure A1 shows it.

Table 4. Synthetic control estimates of the effect on log neonatal mortality

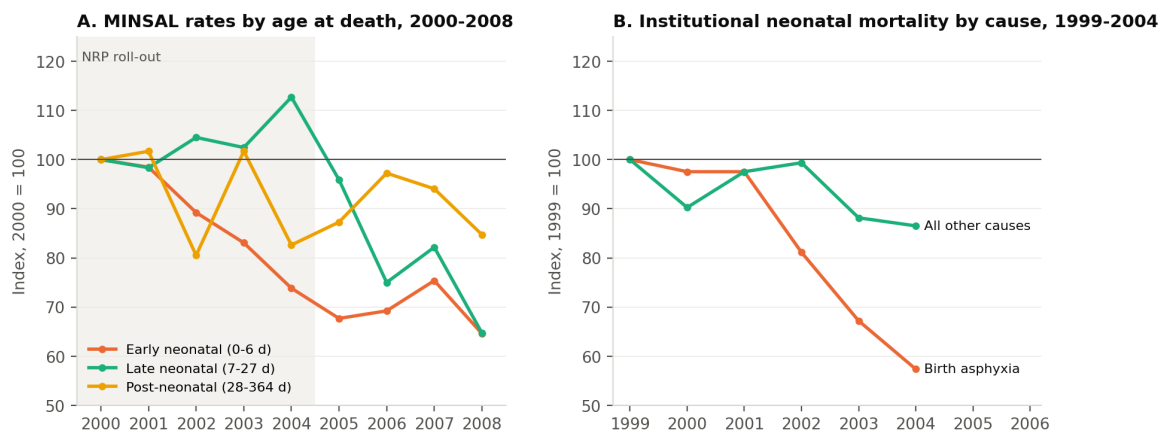
Specification	2000-04	2005-09	2010-19	2000-19	Inference
A. All pre-period lags, log, IGME 2021	-4.0%	-9.9%	-19.4%	-13.4%	RMSPE-ratio rank 9/25 ($p = 0.36$); mean-gap rank 8/25 ($p = 0.32$); good-fit subset $p = 0.60$
B. Covariates and three lags, nested V	-4.0%	-7.4%	-21.9%	-14.2%	.
C. Ridge-augmented (Ben-Michael et al.)	-2.6%	-8.6%	-19.2%	-12.7%	permutation $p = 0.12$
D. Levels, per 1,000	-0.35	.	-1.23	-0.82 (-7.7%)	permutation $p = 0.48$
E. IGME 2024 vintage, 1990-2024	-6.3%	-17.1%	-33.9%	-23.7% (2020-24: -47.2%)	permutation $p = 0.12$
F. Restricted pool (no Haiti, Venezuela)	.	.	.	-14.2%	permutation $p = 0.35$
In-time placebo (fictitious start 1995)	.	.	.	+1.7% (1995-99)	.
Leave-one-out range, specification A	.	.	.	-7.4% to -16.0%	.

Note: Entries are mean post-period gaps converted from log points to percentages (levels for D). Permutation p-values are the rank of El Salvador among the 25 units divided by 25.

The permutation distribution is where the estimate has to be read. Among the 24 placebo countries, seven show mean gaps over 2000-2019 more negative than El Salvador's minus 0.14 log points (Cuba, Bolivia, Honduras, Brazil, Uruguay, Trinidad and Tobago, Peru), and El Salvador ranks 8th on the mean gap and 9th on the RMSPE ratio. Restricting the comparison to the 14 placebos with a pre-period fit within five times El Salvador's, the ratio rank gives $p = 0.60$. The augmented estimator's permutation gives 0.12, as does the 2024 vintage. None of these is below conventional thresholds, and the paper does not claim otherwise. What the exercise establishes is the sign and the rough magnitude of El Salvador's excess decline once its pre-2000 trajectory is matched, and the fact that excess declines of that size occurred in a third of the region's countries without a documented national resuscitation program. The in-time placebo is reassuring about the method rather than about the effect: a fictitious start in 1995 produces a gap of +1.7 percent over 1995-1999.

6.3 Age composition during the roll-out

The Ministry series by age at death carry the most program-specific information in the administrative data, and Figure 4A shows them indexed to 2000. Over the roll-out, 2000 to 2004, the early neonatal rate fell 26 percent, from 6.5 to 4.8 per thousand; the late neonatal rate rose 13 percent, from 4.9 to 5.5; and the post-neonatal rate fell 17 percent. The share of neonatal deaths occurring in the first week fell from 57 to 47 percent. Between 2004 and 2008 the pattern reversed: the early neonatal rate fell a further 12 percent, the late neonatal rate fell 43 percent and the post-neonatal rate was flat. By 2008 the early and late rates had fallen by the same 35 percent from their 2000 values and the first-week share was back at 57 percent.



Panel A: MINSAL series before the 2009 reporting change. Panel B: institutional series reported in Revelo-Castro et al. (PAS 2005); asphyxia fell from 1.22 to 0.70 per 1,000.

Figure 4. Where the roll-out shows up. Panel A: MINSAL rates by age at death, indexed to 2000, before the reporting change. Panel B: institutional neonatal mortality by cause, 1999-2004, from the hospital reporting forms summarised in Revelo-Castro et al. (2005).

Table 5. Changes in institutional mortality by age at death and by cause, El Salvador

Period	Early neonatal (0-6 d)	Late neonatal (7-27 d)	Post-neonatal (28-364 d)	Neonatal (0-27 d)	First-week share of neonatal deaths
2000 to 2004 (roll-out)	-26%	+13%	-17%	-9%	0.57 to 0.47
2004 to 2008	-12%	-43%	+3%	-29%	0.47 to 0.57
2000 to 2008	-35%	-35%	-15%	-35%	0.57 to 0.57
2008 to 2009 (reporting change)	-12%	-76%	-34%	-39%	0.57 to 0.83
1999 to 2004, hospital forms	Asphyxia: -43% (1.22 to 0.70)	All other causes: -13%	All causes: -16% (12.2 to 10.2)	Asphyxia share: 10.0% to 6.9%	

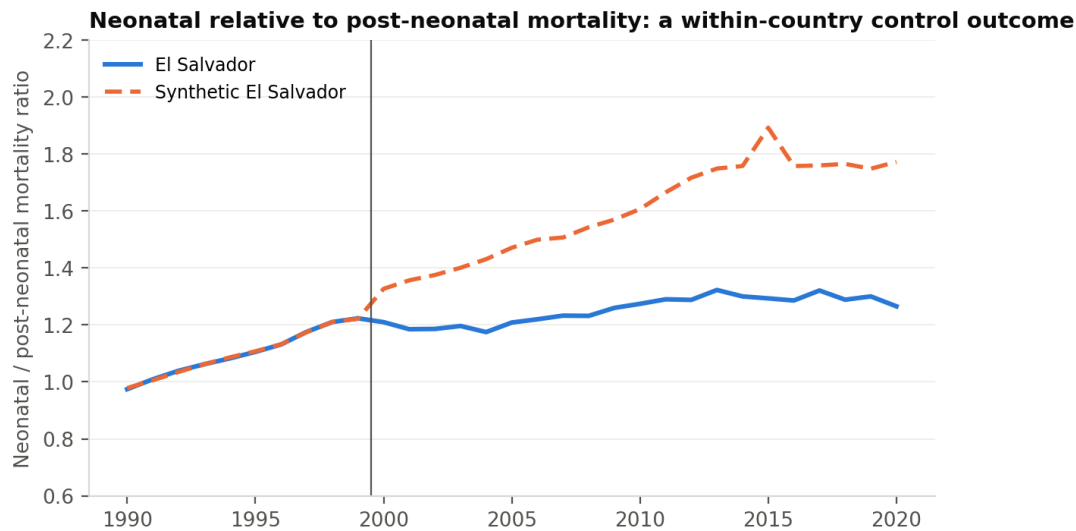
Note: Percent changes in rates per 1,000 live births; MINSAL series for the age groups, Revelo-Castro et al. (2005) for the causes.

This sequence is what a resuscitation program should produce and what a general improvement in newborn care should not. Deaths from intrapartum events happen on the first day or two; a program that prevented some of them would lower the first-week rate, leave the second-to-fourth-week rate alone and shift the composition of neonatal deaths toward later ages. The late neonatal decline that followed in 2004-2008 is consistent with the other components of the Ministry's newborn package, above all neonatal care units and the management of infection, arriving after the resuscitation training rather than with it. The sequence is also consistent with a single-year artefact in a short series, and the paper treats it as a bound rather than an estimate for that reason.

6.4 The control outcome across countries

Figure 5 applies the synthetic control to the log ratio of neonatal to post-neonatal mortality. In the region as a whole neonatal deaths fell more slowly than post-neonatal deaths after 2000, so

the ratio rises in the synthetic control, from 1.2 in 1999 to 1.8 by 2015. In El Salvador it stays flat at about 1.3: neonatal deaths kept pace with the fall in deaths after the first month. The gap averages minus 14 percent over the roll-out, minus 19 percent in 2005-2009 and minus 26 percent in 2010-2019, for minus 21 percent over 2000-2019; El Salvador ranks 5th of 25 on the RMSPE ratio ($p = 0.20$). The weights here are different, Guatemala (0.42) and Brazil (0.34) with small contributions from Belize, Uruguay, Venezuela, Honduras and the Dominican Republic, because the ratio path is a different object from the level path. The raw triple difference points the same way: over 1999-2019 El Salvador's log neonatal-to-post-neonatal ratio rose 0.27 log points less than the donor average (rank 9 of 25); over the pre-period 1990-1999 the corresponding figure was +0.07 (rank 17), so the divergence begins with the program.



UN IGME 2021. Post-neonatal deaths (28-364 days) are not a target of resuscitation training. Synthetic control on the log ratio, 24 Latin American donors.

Figure 5. Neonatal relative to post-neonatal mortality, El Salvador and its synthetic control (UN IGME 2021). Post-neonatal deaths are not a target of resuscitation training; the ratio nets out improvements common to all infant deaths.

Two caveats apply. The UN IGME neonatal and infant series for El Salvador are both model outputs, and their ratio inherits the model's smoothing, so the exact timing of the divergence should not be over-read. And the control outcome nets out country-wide improvements, not improvements specific to newborn care; a neonatal intensive care expansion would move the ratio in the same direction as the program. The result narrows the set of competing explanations without eliminating them.

6.5 Cause of death

The hospital reporting forms give the cleanest cause-specific evidence and the shortest series. Between 1999 and 2004 institutional neonatal mortality fell 16 percent, from 12.2 to 10.2 per thousand; mortality attributed to birth asphyxia fell 43 percent, from 1.22 to 0.70, and mortality from all other causes fell 13 percent, from 10.98 to 9.50 (Figure 4B). Asphyxia accounted for 10 percent of institutional neonatal deaths in 1999 and 7 percent in 2004. The difference between the two cause groups, 0.52 deaths per thousand live births, is the paper's upper bound on the effect of the roll-out, and it is an upper bound because asphyxia deaths were falling before 1999

for other reasons and because reclassification of causes after training is possible in either direction.

The WHO cause-of-death estimates cannot extend this comparison across countries. For El Salvador they show intrapartum-related deaths falling 62 percent between 2000 and 2017 and all other neonatal causes falling 60 percent, with the intrapartum share moving from 18.4 to 17.7 percent; for Nicaragua and Ecuador every cause falls by exactly the same proportion. These are the signatures of modelled fractions applied to a declining envelope, and they carry no information about whether asphyxia deaths fell faster than others in any of these countries. The paper reports them for completeness and does not use them. Section 8 describes the registered-death data that would allow the comparison to be made properly.

6.6 The region as context

Figure 6 places El Salvador among its neighbours. Between 2000 and 2019 its neonatal mortality rate fell 57 percent in the UN IGME 2021 estimates, the largest proportional decline of the 25 countries, just ahead of Peru; the donor average was 37 percent. The rate was also falling fast in the 1990s, 31 percent against a regional average of 22, and the edge persisted in the 2000s (35 against 22) and the 2010s (30 against 20). This is why the paper compares El Salvador with a synthetic control built to match its 1990s path rather than with the regional mean: on a regional-mean comparison the excess decline after 2000 would be about twice as large as the synthetic control estimate, and half of the difference would be El Salvador's pre-existing trajectory.

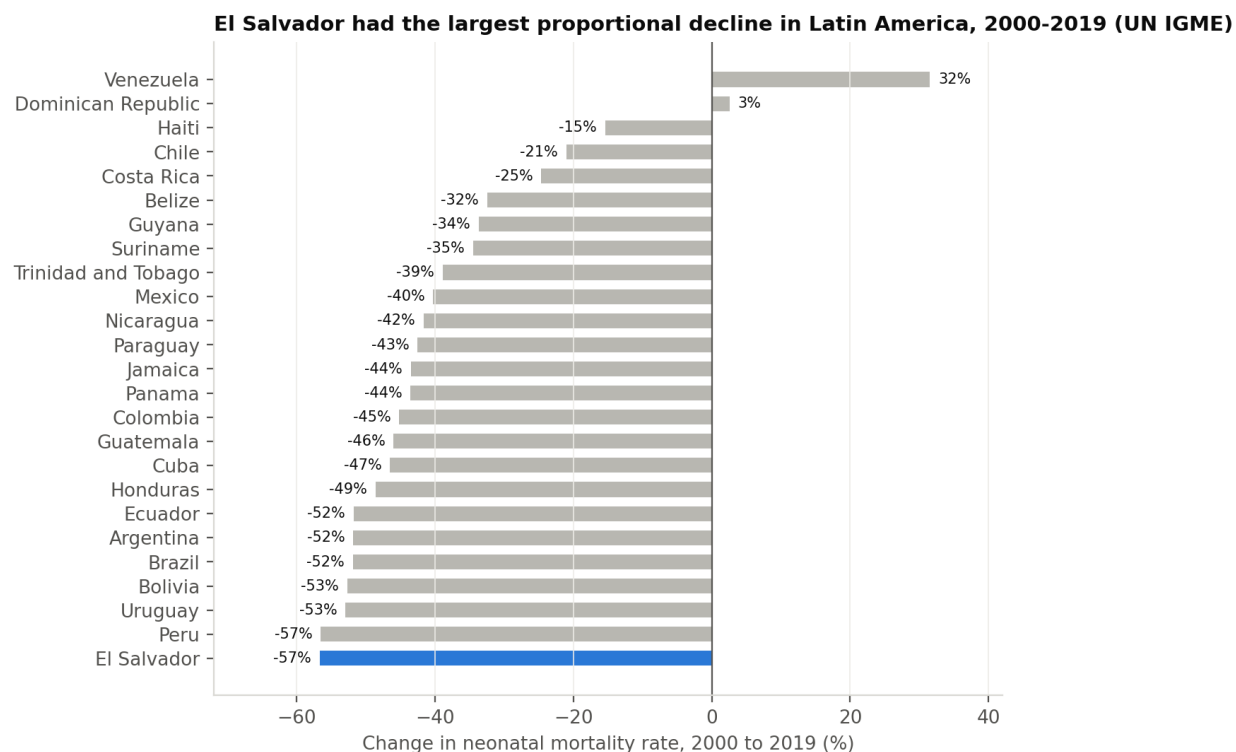


Figure 6. Change in the neonatal mortality rate, 2000 to 2019, 25 Latin American and Caribbean countries (UN IGME 2021).

7. Bounds and cost-effectiveness

The evidence assembled above does not yield a point estimate with a defensible confidence interval, and the paper does not manufacture one. It yields bounds, and the bounds are enough to answer the policy question.

The upper bound comes from the cause composition of the roll-out years: 0.52 deaths per thousand institutional live births, the fall in asphyxia mortality in excess of the fall in other causes between 1999 and 2004. A literature-consistent central value applies the 30 percent reduction in intrapartum-related deaths from the Lives Saved Tool review to El Salvador's 1999 asphyxia rate, which gives 0.37 per thousand. The national synthetic control, taken at face value, implies about 1.8 deaths per thousand by 2019, but that figure includes every change in newborn care in El Salvador after 2000 and its permutation rank says it could be noise; it is a ceiling on the sum of all newborn-care improvements, not a bound on the program.

Translating the bounds into deaths requires the number of births the trained staff attended. El Salvador registered about 125,000 births in 2004 and about 105,000 by the late 2010s; with institutional delivery between 70 and 92 percent in the early 2000s, the covered population was roughly 90,000 to 115,000 births a year. The upper bound then corresponds to 47 to 60 deaths averted per year, the literature value to 33 to 42, and one tenth of the upper bound to 5 or 6.

Table 6. Cost per death and per DALY averted under alternative effect sizes and unit costs

Effect size	Deaths averted per year	Cost per death averted (US) <i>Cost per DALY averted (US)</i>	Maximum, as share of GDP per capita	
Upper bound (0.52 per 1,000)	47 to 60	2,100 to 8,200	72 to 275	0.07
Literature (30% of 1.22 per 1,000)	33 to 42	3,100 to 11,700	102 to 390	0.09
Half the literature value	16 to 21	6,100 to 23,400	203 to 780	0.19
One tenth of the upper bound	5 to 6	21,500 to 82,400	716 to 2,745	0.66

Note: Annual program cost between US\$128,000 and US\$385,000, from 885 trainings a year at US\$100 to 300 per participant-course plus US\$40,000 to 120,000 of fixed costs (committee, coordination, equipment upkeep). Thirty discounted DALYs per neonatal death averted. GDP per capita of El Salvador in 2019: US\$4,143 (constant dollars). Unit costs are illustrative and drawn from the range reported for NRP and Helping Babies Breathe courses in low- and middle-income settings (Vossius et al. 2014; Chaudhury et al. 2016); program cost records would replace them.

On the cost side the program was cheap. It delivered 885 trainings a year on average; at all-in unit costs of 100 to 300 dollars per participant, plus fixed costs of the committee, coordination and delivery-room equipment, the annual budget lies between about 130,000 and 385,000 dollars. Table 6 combines the two sides. At the upper bound the cost per death averted is between 2,100 and 8,200 dollars and the cost per DALY between 72 and 275 dollars, under a tenth of GDP per capita. At one tenth of the upper bound, five or six deaths a year, the cost per DALY rises to between 716 and 2,745 dollars, still below the one-times-GDP-per-capita

threshold that the region's health technology assessments treat as cost-effective, and far below the willingness-to-pay implied by El Salvador's other health spending. The conclusion does not depend on which of the bounds is right. It would take an effect an order of magnitude smaller than the smallest figure in the table, or costs several times higher than any reported for these courses, to make the program a poor use of money. That is a stronger policy statement than any point estimate the data could support, and it is the reason the paper is comfortable with bounds.

8. What would settle the question

The scientific question, how many deaths the program prevented, is answerable with data that exist or existed. Three sources, in order of value, would each transform the analysis.

A hospital-level panel. The 2010 and 2020 protocols describe the data the study was designed to collect: for each of the 28 hospitals, six-month periods before and after the hospital's training, with all neonatal deaths by cause and by 250-gram birth-weight group. Because the roll-out took four years and reached hospitals in sequence, that structure supports a staggered difference-in-differences with hospital and period fixed effects, an event-study around each hospital's training date, and the modern estimators that handle staggered adoption with heterogeneous effects (Callaway and Sant'Anna 2021; Sun and Abraham 2021; de Chaisemartin and D'Haultfoeulle 2020). Within-country variation in timing is what the national series lacks, and it removes the need for cross-country comparison altogether. The birth-weight groups would add a second test: resuscitation should matter most for term and near-term infants, and least for those under 1,000 grams, whose deaths are dominated by prematurity.

Vital registration by cause and age in days. El Salvador reports its registered deaths to the WHO Mortality Database by ICD-10 code and age group, including deaths at 0 to 6 and 7 to 27 days, for most years since 1997. That file allows the cause-specific comparison of Section 6.5 to be made nationally and annually: intrapartum hypoxia and birth asphyxia (P20-P21) against causes the program does not touch, such as congenital malformations (Q00-Q99) and the complications of extreme prematurity (P07), before and after 2000 and relative to countries in the same database. Completeness of registration is itself a concern in El Salvador, and it changes over time, but the ratio of treated to untreated causes is robust to completeness that is common across causes. This is the design of Jayachandran, Lleras-Muney and Smith (2010) applied to newborns, and it is available at no cost.

Birth histories. The FESAL surveys of 1998, 2002-03 and 2008 and the 2014 MICS collected complete birth histories with the place of delivery and a household wealth index. Neonatal mortality among facility births relative to home births, before and after 2000, is a third difference that the program should produce and that nothing else in the reform period should, and it can be estimated from public microdata. The same surveys answer the distributional question that the aggregate data cannot: whether the decline in neonatal mortality after 2000 was shared by the poorest quintiles and by rural households, whether the gap between facility and home births widened during the roll-out and closed as institutional delivery approached universality after 2009, and who, if anyone, was left outside the room.

Program cost records, which the Committee and the SALSA project must once have held, would replace the illustrative unit costs of Section 7 with actual ones. None of these sources requires new fieldwork.

9. Conclusion

Few countries have documented what El Salvador's Ministry of Health did: it trained the staff of every public delivery hospital in a standard neonatal resuscitation curriculum in four years and kept training them for fifteen more. Over the same period the country's neonatal mortality fell faster than anywhere else in Latin America. The temptation to connect the two with a before-and-after regression is understandable and, this paper has argued, mistaken, not because the program did not work but because the data the temptation relies on cannot show that it did. The administrative series was falling before the program and did not bend at its start; the break the earlier analysis found in 2009 is a change in how deaths were counted.

What the data can show is more modest and more interesting. Against a synthetic comparator built from countries that matched its 1990s trajectory, El Salvador's neonatal mortality fell 13 to 24 percent more than expected over two decades, and its neonatal deaths kept pace with post-neonatal deaths when the region's did not; both patterns are consistent with a program effect and with the other improvements in newborn care in El Salvador after 2000, and neither is distinguishable from the region's ordinary variation. During the roll-out itself, deaths fell where a resuscitation program would make them fall, in the first week and among infants recorded as asphyxiated, and not elsewhere. That evidence supports a bound of about half a death per thousand institutional births, roughly fifty deaths a year at the time, and the bound is enough to establish that the program was cost-effective by any threshold in use in the region even if its true effect was a fraction of that.

One question the paper raises and cannot answer deserves to be stated on its own. The program was universal within the public hospitals and reached the households that depend on them, which are the households at highest risk; in that sense it was an instrument of equity as much as of survival. But its design stopped at the delivery-room door, and for the births that took place outside a facility during the roll-out, the poorest and the most remote, the program did nothing until institutional delivery caught up with them a decade later. Whether the decline documented here was shared across wealth quintiles and regions, or concentrated among those already inside the system, is a question for the birth histories in the household surveys, and it should be asked before the program is held up as a model.

The larger lesson concerns evaluation. Programs of this kind will continue to be assessed from one national series and a start date, and the assessment will continue to look persuasive whenever the intervention is one that clinical evidence says should work. The remedy is not to distrust the clinical evidence. It is to ask, before the regression is run, what the counterfactual is, what else changed at the same time, whether the data system changed, and which deaths the intervention could and could not have prevented. Where those questions can be answered from the structure of the intervention, as they partly can here, aggregate data can still say something. Where they cannot, the hospital-level records that the program's designers planned to collect,

and the country's own registration of deaths by cause and age, are the way to find out how many first breaths the training actually bought.

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Appendix

A. Structural breaks in the administrative series

The 2023 manuscript located structural breaks with supremum Wald tests on regressions of each log mortality rate on the log of providers trained. Table A1 re-runs the test on the rates themselves, for an unknown single break in level and trend with 15 percent trimming (Andrews 1993). Every rate except the early neonatal one breaks in 2009, the year of the reporting change; the early neonatal rate, which is insensitive to the capture of deaths after discharge, breaks in 2014 with a much smaller statistic. In the manuscript, the early neonatal break was reported at 2013 with a p-value of 0.81.

Table A1. Supremum Wald tests for an unknown break, log mortality rates (MINSAL)

Rate	Window	Observations	Estimated break	Sup-Wald
Neonatal (0-27 d)	1995-2020	26	2009	112.9
Early neonatal (0-6 d)	2000-2020	21	2014	41.6
Infant (0-364 d)	2000-2020	21	2009	84.0
Late neonatal (7-27 d)	2000-2020	21	2009	114.0
Post-neonatal (28-364 d)	2000-2020	21	2009	27.8

Note: Break in level and trend, 15 percent trimming at each end. The largest one-year drop in the neonatal series is 2009 (-39%).

B. Synthetic control details

Table B1. Donor weights, specification A (all pre-period lags, log NMR, UN IGME 2021)

Donor	Weight
Peru	0.514
Guyana	0.193
Chile	0.131
Honduras	0.081
Belize	0.060
Venezuela	0.020

Table B2. Predictor balance, specification B (covariates and three lags, nested V)

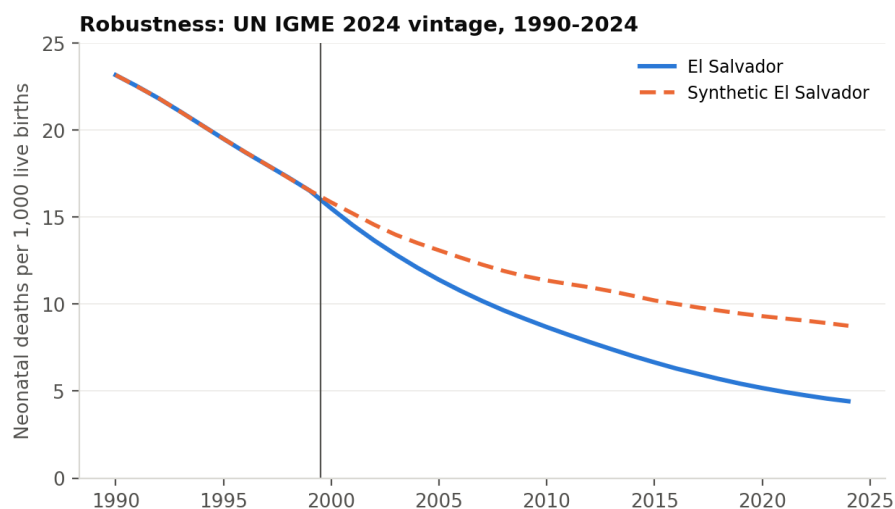
Predictor	El Salvador	Synthetic	V weight
Log NMR 1990	3.13	3.12	0.00
Log NMR 1995	2.94	2.94	0.00
Log NMR 1999	2.77	2.77	1.00
Log GDP per capita (1990s mean)	7.89	8.21	0.00
Urban share, % (1990s mean)	52.16	52.16	0.00
Mean years of schooling, women 15-44	5.92	7.54	0.00
Births attended by skilled staff, %	88.90	80.36	0.00
Adolescent fertility, per 1,000	108.96	95.32	0.00
Under-five mortality, per 1,000 (1990s mean)	48.97	47.46	0.00

Note: The nested optimisation places all weight on the 1999 outcome; the covariate columns describe the comparator that results rather than targets that were matched.

Table B3. Placebo distribution, specification A: every unit treated in turn as if it were El Salvador

Unit	Pre-period RMSPE	Mean gap 2000-2019 (log)	Mean gap (%)	RMSPE ratio
Cuba	0.2160	-0.559	-43%	2.7
Bolivia	0.0650	-0.281	-25%	5.5
Honduras	0.0052	-0.280	-24%	66.5
Brazil	0.0004	-0.205	-19%	514.2
Uruguay	0.0076	-0.188	-17%	33.9
Trinidad and Tobago	0.0013	-0.172	-16%	179.1
Peru	0.0470	-0.160	-15%	4.2
El Salvador	0.0019	-0.144	-13%	91.6
Paraguay	0.0007	-0.093	-9%	159.7
Jamaica	0.0026	-0.088	-8%	36.1
Panama	0.0007	-0.072	-7%	127.9
Guatemala	0.0004	-0.056	-5%	240.3
Ecuador	0.0071	-0.034	-3%	17.3
Costa Rica	0.0534	-0.021	-2%	1.8
Colombia	0.0011	-0.010	-1%	63.6
Mexico	0.0010	+0.003	+0%	127.4
Guyana	0.0205	+0.003	+0%	0.5
Argentina	0.0166	+0.040	+4%	6.4
Belize	0.0071	+0.085	+9%	21.8
Suriname	0.0009	+0.100	+11%	118.8
Nicaragua	0.0002	+0.143	+15%	955.6
Chile	0.0511	+0.182	+20%	4.2
Haiti	0.0146	+0.265	+30%	21.9
Dominican Republic	0.0225	+0.282	+33%	15.0
Venezuela	0.0352	+0.364	+44%	14.6

Note: Sorted by the mean post-period gap. El Salvador ranks 8 of 25 on the mean gap and 9 of 25 on the RMSPE ratio. Units with pre-period RMSPE above five times El Salvador's (0.0093) are excluded from the good-fit subset used for the restricted permutation test.



Mean gap 2000-2019: -24%; permutation $p = 0.12$. Donor weights: Peru 0.30, Belize 0.21, Paraguay 0.18, Guyana 0.12, Ecuador 0.12, Chile 0.06, Venezuela 0.01

Figure A1. Synthetic control on the UN IGME 2024 vintage, 1990-2024.

C. Control outcome and descriptive dose-response

Table C1. Donor weights, synthetic control on $\log(\text{NMR} / \text{post-neonatal MR})$, UN IGME 2021

Donor	Weight
Guatemala	0.416
Brazil	0.336
Belize	0.068
Uruguay	0.059
Venezuela	0.057
Honduras	0.046
Dominican Republic	0.018

Note: Pre-period RMSPE 0.0025; mean gap 2000-2019 -21%; rank 5 of 25 on the RMSPE ratio ($p = 0.20$).

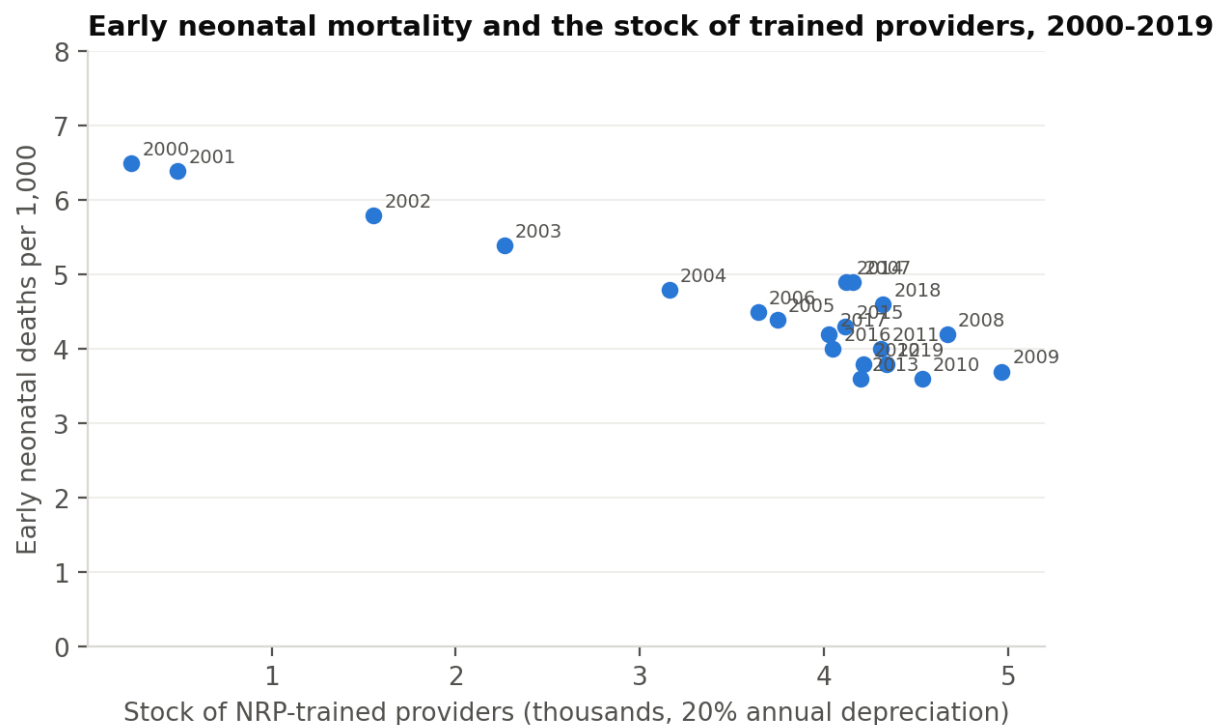


Figure A2. Early neonatal mortality and the depreciated stock of trained providers, 2000-2019 (descriptive).

The scatter in Figure A2 shows why no dose-response estimate is reported: the stock rises to its plateau by 2004 and varies little thereafter, so the relationship is a comparison of the roll-out years with everything after, which is the before-and-after comparison in another form.

D. Cost-effectiveness inputs

Annual trainings, mean 2000-2019: 885. Annual program cost under the assumptions of Table 6: US\$128,475 to US\$385,425. DALYs per neonatal death averted: 30 (3 percent discount rate, life expectancy of about 73 years). GDP per capita, 2019: US\$4,143 (constant dollars, World Development Indicators via Gapminder). The full grid of scenarios is in [results/costeff_table.csv](#).

E. Code and data

Every number in the paper is produced by seven scripts (data build, interrupted time series, synthetic control, composition and control outcomes, figures, cost-effectiveness, and this document) from the Ministry spreadsheets and the public cross-country files; the provenance file lists every data decision. The Ministry series are aggregate national rates and contain no individual records. Their release as a file is subject to the Ministry's and the study team's agreement; the cross-country panels are public.